Fiscal Year:	FY 2025	**Task Last Updated:**	FY 08/08/2025
PI Name:	Rithidech, Kanokporn Ph.D.		
Project Title:	Effects of Space Flights on the Proteome of Astronauts' Plasma		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Stony Brook	**State:**	NY
Zip Code:	11794-8691	**Congressional District:**	1
Comments:			
Project Type:		**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	04/01/2016	**End Date:**	12/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	Flight definition NOTE: End date changed to 12/31/2025 per JSC Grants Office (Ed., 8/5/25). NOTE: End date changed to 12/31/2024 per NSSC information (Ed., 9/2/24). NOTE: End date changed to 05/31/2024 per T. Sirmons/JSC (Ed., 5/16/23). NOTE: End date changed to 12/31/2023 per PI and NSSC information (Ed., 3/21/23). NOTE: End date changed to 04/26/2023 per T. Sirmons/JSC (Ed., 3/3/23). NOTE: End date changed to 12/31/2022 per NSSC information (Ed., 1/4/22) NOTE: End date changed to 12/31/2021 per NSSC information (Ed., 12/31/20) NOTE: End date changed to 12/31/2020 per NSSC information (Ed., 6/12/20) NOTE: End date changed to 3/31/2020 per NSSC information (Ed., 3/25/19)		
Key Personnel Changes/Previous PI:	No change		

COI Name (Institution):	Martin, Dwight Ph.D. (State University of New York, Stony Brook)
Grant/Contract No.:	NNX16AH80G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	NOTE: This is an integrated project consisting of Dr. Brian Crucian's "Functional Immune Alterations, Latent Herpesvirus Reactivation, Physiological Stress, and Clinical Incidence Onboard the International Space Station" directed research; and Dr. Richard Simpson's "The Impact of an ISS Mission on the Anti-Viral and Functional Properties of NK-cells, T-cells, B-cells and Dendritic Cells," Dr. Kanokporn Rithidech's "Effects of Space Flights on the Proteome of Astronauts' Plasma," and Dr. Honglu Wu's "DNA Damage in the ISS Astronaut's Lymphocytes and Their Association with Stress-Induced Immune Dysfunction" solicited research. Space flight results in exposure of astronauts to several stressors, such as space radiation, microgravity, and physiological stress, that could exacerbate the risks of adverse health effects. To protect astronauts, we must improve our understanding of molecular changes that influence immunological conditions associated with increased astronaut health risks. The in vivo response to the space environment is complex, involving multiple proteins associated with various signal transduction cascades, resulting in different outcomes. Molecular mechanisms responsible for such diverse consequences are poorly understood. It is, therefore, essential to characterize the protein signatures of responses to the space environment in blood plasma samples from astronauts, collected at pre-, in-, and post-flights. Such analyses should help to reveal a particular set of proteins causing adverse immunological changes and to develop methods that help to prevent, or at least to counteract, these effects. In this flight definition project, we will use cutting age proteomic technology to determine protein alterations, qualitatively and quantitatively, in plasma samples collected from astronauts before, during, and after space flights. Our findings will help to provide an understanding of the time course and etiology of immune changes induced by the space environment. Furthermore, since pre- and post-flight samples, in addition to the in-flight samples, will be evaluated in the same astronaut, the direct effects of the space environment can be determined. Hence, our findings will provide high-priority and highly relevant information to NASA. We will further correlate protein expression profiles to the available data on immune dysfunction detected in each astronaut. This approach makes it possible to determine potential predictive biomarkers for space-flight-induced immune dysregulation. Consequently, effective countermeasures against such harmful effects of the space environment can be identified.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The data obtained from our studies will provide crucial insights into the timeline of immune changes induced by the space environment. Our findings will be highly relevant to NASA, offering priority information that can enhance understanding of these effects. We will compare protein expression profiles with available data on immune dysfunction observed in astronauts. This approach allows us to identify potential predictive biomarkers for immune dysregulation caused by space flights. Such knowledge is vital for assessing health risks and will aid in developing effective countermeasures to support astronauts, space travelers, and individuals on Earth with immune dysfunction.
	We identified 693 non-redundant proteins (at > 99% confidence). Statistical analysis of all 693 quantified proteins was performed, and proteins with an ANOVA p-value of ≤ 0.05 were selected for downstream analysis. Among the 693 identified proteins, we found that the expression levels of 16 proteins are significantly altered (i.e., increased or decreased) during spaceflight. The number of unique and shared identified proteins in the control and each time point ranged from 249 to 320 proteins. As expected, we observed a good overlap among the proteins quantified at each time point, in which 241 proteins were common or shared at all time points. Due to the individual variance in protein abundance between sampled astronauts, we decided to filter for proteins that were quantified in at least three of the sampled astronauts at least one time point. Such stringent criteria facilitate the identification of proteins that may be directly regulated in response to the space environment and the re-adaptation of the Earth's gravity. The distribution of log-transformed Label-Free Quantification (LFQ) values of the proteins identified at different time points was then analyzed by violin plots, which show that the identified proteins in the R+30 are more dispersed than in other groups. The biological functions of the proteins that were significantly altered in expression were classified into 10 categories: acute inflammatory response, actin skeleton organization, blood coagulation, defense response to other organisms, extracellular matrix organization, negative regulation of endopeptidase activity (metabolic process), phagocytosis, platelet degranulation, positive regulation of cytokine secretion, and tissue homeostasis. Some proteins have multiple biological functions, e.g., APOL1, SAA1, and TE. Following the pre-flight, the levels of certain proteins, such as Apolipoprotein L1 (APOL1) and inter-alpha-trypsin inhibitor heavy chain H2 (ITIH2), returned to or close to their pre-flight levels. Meanwhile, the levels of some proteins involved in the actin cytoskeleton (e.g., pleckstrin or PLEK) and coagulation (e.g., platelet glycoprotein 1b alpha chain or GP1BA) decreased postflight. Only two proteins increased postflight as compared to the preflight level, i.e., Brain acid soluble protein 1 (BASP1) and Insulin-like growth factor-binding protein 4 (IGFBP4). Although the mechanisms behind the downregulation and upregulation of these proteins remain unclear, they may have important roles in response to spaceflight or during the process of readjustment to Earth. This could potentially impact cellular and tissue integrity, as well as homeostasis, leading to long-term health risks. We created a heatmap and the dendrogram of these 16 proteins, comparing their expressions and relative changes (not absolute values) of expression of eight astronauts collected preflight, in-flight, and post-flight. Circle size and color represent the Z-score. Each row represents a temporal expression of a protein. Each column represents a time point of sample collection. In the heatmap, red color represents high expression levels of proteins, and green represents very low expression levels. Changes in the expression level (either increased or decreased) of each protein were first detected in the in-flight samples (early responses). We identified 693 non-redundant proteins and observed significant expression changes in 16 of them, first noted in in-flight samples. This indicates that the space environment affects physiological systems. The expression levels of these

Task Progress:	proteins at day 90 post-flight (R+90) can be categorized into three groups: Group A: Seven proteins exhibited similar or minimal changes (or slight increases or decreases) in expression levels between pre-flight and post-flight. This group includes APOL1, PEPD, THBS4, ITIH2, SERPIND1, SAA1, TNXB, and TF; Group B: Seven proteins displayed lower expression levels in postflight samples compared to preflight samples. This group includes GP1BA, Q9UL85, PGD, CPB2, CRP, PLEK, and TF; Group C: Two proteins showed higher expression levels in postflight samples compared to preflight samples, which are BASP1 and IGFBP. Each group consists of proteins with diverse functions, as reflected in the sub-cluster classifications. Our data suggests that proteins in Group A were able to readapt to Earth's gravity 90 days after returning, although it is still unclear how this return to near preflight levels occurred. It is possible that the repair systems played a role. It remains unknown whether the levels of these proteins will remain consistent over time post-flight. Several proteins with altered expression were previously reported in the NASA Twin Study or in the Space Omics and Medical Atlas (SOMA) and the international astronaut biobank. However, some proteins, such as BASP1 and Q9UL85, were not reported in either study, while others, like PEPD and THBS4, were detected only in the Twin Study and not in the SOMA databank. It is important to note that in-flight samples were not included in the proteomic analyses in the SOMA databank; however, our study and the Twin Study did include such samples for proteomic analyses. Our results show that several immune-related proteins (e.g., CPB2, CRP, TF, and Q9UL85) had significantly reduced expression levels at R+90, indicating that the space environment profoundly affects the immune system. This reduction, first noted during the flight, persisted in post-flight samples and suggests an imbalance in astronauts' immune systems, which has been linked to immune dysfunctions like altered cytokine production, phagocytosis, and T cell signaling. These issues may increase chronic inflammation and health risks for crewmembers. The reduced expression level of the PLEK protein in the R+90 post-flight samples, compared to pre-flight levels, indicates that the space environment may downregulate actin-binding proteins. Microgravity and space radiation might lead to DNA methylation of the genes encoding these proteins, resulting in decreased expression. It remains unclear if these low protein levels will persist or return to pre-flight levels. Notably, we observed changes in SERPIN family proteins, particularly SERPIND1 (Heparin cofactor II), which is important for coagulation. Disruption of vascular homeostasis poses a significant health risk for astronauts. Our previous studies also noted a decrease in SERPINC1, another SERPIN family member related to coagulation. Thus, the suppression of specific SERPIN proteins may indicate spaceflight-induced coagulation dysfunction and could inform therapeutic strategies. A significant finding is the increased levels of BASP1 (brain-abundant, membrane-attached signal protein 1) and IGFBP4 (insulin-like growth factor binding protein 4) in the R+90 samples. BASP1 is crucial for signal processing, synaptic plasticity, and neurite outgrowth, being associated with the cell membrane and cytoskeleton. BASP1 is found in the brain as well as in other tissues, including the kidney, lymphoid tissues, neck, prostate, and epididymis. The observed increase in plasma BASP1 levels after space missions suggests a potential link to tissue injuries from spaceflight, warranting further investigation. Elevated BASP1 levels have been noted in various solid tumors. Such findings indicate its potential significance in carcinogenesis. While its exact role in this context remains unclear, BASP1 may have a prognostic value in several types of cancer. However, a tumor-suppressor role of BASP1 was recently reported in MYC-related breast cancer. The finding of increased IGFBP4 levels is important because this protein is associated with senescence. Exposure to toxic agents such as radiation (X-rays) has been found to enhance the level of IGFBP4 and trigger its release into the bloodstream. Senescence is associated with impairments in the immune, cardiovascular, and nervous systems, as well as increased cancer risk in astronauts. While the mechanisms of IGFBP4 upregulation are unclear, the spaceflight environment may contribute to its overexpression. The increased IGFBP4 levels found in astronauts' plasma could serve as a biomarker for space stressors and aid in developing biological countermeasures.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Articles in Other Journals or Periodicals	Rithidech K, Ferrier RM, Aryal U, Peanlikhit T, Crucian B. "Effects of the space environment and re-adaptation to Earth's gravity on astronauts' plasma proteome. " Life Sci Space Res, In revision, as of August, 2025. , Aug-2025
Articles in Peer-reviewed Journals	Rithidech K, Mohallem R, Aryal UK, Peanlikhit T, Crucian B. "Effects of the space environment and re-adaptation to Earth's gravity on astronauts' plasma proteome." Life Sci Space Res. 2026 Jan;48:78-87. https://doi.org/10.1016/j.lssr.2025.09.001 , Jan-2026